Data Dictionary

Telegram Backed Hybrid Cloud Storage Management System

Version 1.0 | Companion to SRS v1.0, Data Flow Diagrams (Level 0, 1, 2) and Decision Table

1. Introduction

This data dictionary defines every data store, data flow, data structure, data element, external entity and process used in the Telegram Backed Hybrid Cloud Storage Management System. The entries are taken directly from the Level 0, Level 1 and Level 2 Data Flow Diagrams, and are cross-checked against the Software Requirements Specification (FR, NFR and BR identifiers) and the Decision Table rules (R1 to R29).

The system is a proxy-based cloud storage application. Users manage a virtual folder tree through a web dashboard, file binaries are streamed to a private Telegram channel through the Telegram Bot API, and a local SQL database (store D1) holds the metadata that maps Users, Folders and Files to Telegram file identifiers.

Where the source documents leave a value open (for example password hashing, JWT algorithm, or the administrator role model), the entry is marked "proposed" or "TBD" and is listed again in Section 9.

2. Notation Used

Symbol	Meaning	Example
`=`	is composed of	`Credentials = Username + Password`
`+`	and (sequence)	`Username + Password`
`[ | ]`	either / or (selection)	`Transfer_Mode = [Standard | Streaming]`
`( )`	optional element	`Error_Message in Upload_Status`
`{ }`	iteration (zero or more repeats)	`Dashboard_View contains {Folder_Record}`
`* *`	comment	`* 0 to 100 *`
`PK / FK`	primary key / foreign key	`User_ID is PK of Users and FK in Folders`

3. Data Store Dictionary

Store	Name	Description
D1	SQL Metadata Database	The single relational store used by processes 1.0, 2.0, 4.0, 5.0, 6.0 and 7.0 (and 3.1 for folder checks). Holds the logical hierarchy Users, Folders, Files. It holds metadata only; no file binaries are stored locally (FR-015, NFR-006). Loss of D1 removes the mapping to remote Telegram files (BR-006).

Composition of D1:

```
D1 = {Users} + {Folders} + {Files}

Users = User_ID + Username + Password_Hash + Role + Created_At
Folders = Folder_ID + User_ID + Parent_Folder_ID + Folder_Name + Created_At
Files = File_Record_ID + User_ID + Folder_ID + Telegram_File_ID + File_Name + File_Size + Upload_Date
```

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3.1 Table: Users

Stores registered accounts. Supports FR-001 to FR-004, NFR-009, BR-001. Written by process 1.0.

Attribute	Data Type	Key	Null?	Description / Constraint
User_ID	INTEGER (auto)	PK	NOT NULL	System generated unique identity of the user.
Username	VARCHAR(50)	UNIQUE	NOT NULL	Login name. Must be unique across the system (FR-002, BR-001). Length 3 to 50 (proposed).
Password_Hash	VARCHAR(255)		NOT NULL	One-way hash of the password; the plain password is never stored. Algorithm is TBD-006 (proposed: bcrypt or Argon2).
Role	VARCHAR(10)		NOT NULL	Allowed values: "user" or "admin". Default "user". Used to authorize the admin export (FR-034). Role management is TBD-009 (proposed).
Created_At	DATETIME		NOT NULL	Timestamp of account creation (proposed).

3.2 Table: Folders

Stores the virtual folder tree. Supports FR-008 to FR-012, BR-002, BR-003. Written and read by process 2.0.

Attribute	Data Type	Key	Null?	Description / Constraint
Folder_ID	INTEGER (auto)	PK	NOT NULL	Unique identity of the virtual folder.
User_ID	INTEGER	FK	NOT NULL	References Users.User_ID. Owner of the folder; every query filters on this (NFR-012).
Parent_Folder_ID	INTEGER	FK	NULL	References Folders.Folder_ID. NULL means the folder is at root level. Gives the recursive hierarchy (FR-010).
Folder_Name	VARCHAR(255)		NOT NULL	Custom name chosen by the user. Should be unique among siblings of the same parent (proposed).
Created_At	DATETIME		NOT NULL	Timestamp of folder creation (proposed).

3.3 Table: Files

Stores one metadata row per uploaded file. Supports FR-020 to FR-028, FR-029 to FR-031, BR-004, BR-006. Written by process 4.0, read by 2.0, 5.0 and 6.0.

Attribute	Data Type	Key	Null?	Description / Constraint
File_Record_ID	INTEGER (auto)	PK	NOT NULL	Internal identity of the file record. This is the only file identifier ever shown to the browser.
User_ID	INTEGER	FK	NOT NULL	References Users.User_ID. Used by the ownership check (Decision Table R21).
Folder_ID	INTEGER	FK	NOT NULL	References Folders.Folder_ID. The virtual folder that contains the file.
Telegram_File_ID	VARCHAR(255)		NOT NULL	Persistent file_id returned by the Telegram Bot API. Protected backend value; never exposed to the browser (NFR-013, R21).
File_Name	VARCHAR(255)		NOT NULL	Original name of the uploaded file. Searched by process 6.0.
File_Size	BIGINT		NOT NULL	Size in bytes. Range 0 to 2 GB (FR-017).
Upload_Date	DATETIME		NOT NULL	Time the metadata was committed (proposed).

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3.4 Relationships and Integrity Rules

- Users 1 : N Folders, and Users 1 : N Files (user isolation, BR-002).
- Folders 1 : N Files, and Folders 1 : N Folders through Parent_Folder_ID (BR-003).
- A Files row is created only after a Telegram_File_ID has been received and the transfer succeeded; this is what makes the file visible (FR-023, BR-004, R11, R12, R17, R18).
- A folder can be deleted only when it has no child folders and no files (FR-012, R23, R24).
- Deleting a Files row only unlinks the file from the virtual tree; the remote Telegram object is not removed (BR-006, R22).

4. External Entity Dictionary

Entity	Alias	Description / Data exchanged
Storage User	System user, web browser	Authenticated end user. Sends: Credentials, Folder_Command, File_Action, Binary_Chunk, Search_Request, Download_Request. Receives: JWT_Token, Dashboard_View, Upload_Status, Streamed_File, errors.
System Admin	Administrator (privileged)	Privileged operator. Sends: Export_Request with Admin_Credentials. Receives: Backup_File.
Telegram Bot API	Private channel	External remote storage service. Receives: Binary_Chunk blocks and File_ID_Request. Returns: Telegram_File_ID, HTTP_429_Response, API_Status, Binary_Packet.

5. Process Dictionary

No.	Process	Purpose	Related FR / Rules
1.0	User auth management	Registers users, validates credentials, issues JWT, blocks unauthenticated access.	FR-001 to FR-005; R1 to R4
2.0	Dashboard & directory	Shows current folder, creates, opens, nests and deletes folders, goes back to parent.	FR-006 to FR-012; R5 to R10, R23, R24
3.0	Streamed upload proxy	Receives chunks, streams them to Telegram, handles 429 and errors, reports progress.	FR-013 to FR-019, FR-022; R11 to R18
3.1	Validate & size broker	Checks folder and file size, chooses standard (up to 50 MB) or streaming mode, rejects files over 2 GB.	FR-016, FR-017; R11 to R13
3.2	Memory buffer streamer	Forwards multipart blocks without holding the full file in RAM or on disk.	FR-014, FR-015; NFR-001, NFR-002
3.3	Rate-limit interceptor	Sends blocks to Telegram, captures file_id, pauses on HTTP 429, reports connection errors.	FR-019, FR-020, FR-022; R14 to R16
3.4	Progress broadcaster	Computes and sends upload percentage to the user.	FR-018; R11
4.0	Metadata indexing	Writes Telegram_File_ID, name, size and folder links into D1, then triggers a dashboard refresh.	FR-021, FR-023; R17, R18
5.0	Secure download	Looks up metadata, verifies ownership, fetches from Telegram and streams to the browser.	FR-024 to FR-026; R19 to R21
5.1	Ownership checker	Verifies the JWT user owns the requested file; denies access otherwise.	NFR-008, NFR-012; R20, R21
5.2	Relational key retriever	Reads Telegram_File_ID from D1 for the authorized file.	FR-025
5.3	Response stream proxy	Requests the file from Telegram and streams binary packets to the browser.	FR-026; NFR-003

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No.	Process	Purpose	Related FR / Rules
6.0	Workspace search	Searches the user's file metadata by name; also receives UI preferences.	FR-029 to FR-032; R25, R26, R29
7.0	Admin operations	Verifies admin permission and exports the SQL metadata backup.	FR-033 to FR-035; R27, R28

6. Data Flow Dictionary

Level column: L0, L1, L2 show the DFD level where the flow appears. Names in the Alias column are the labels written on the diagrams.

Data Flow	Alias	Composition	Source → Destination	Level
Credentials	Registration, login; Registration_Data	Username + Password	Storage User → 1.0	L0, L1
JWT_Token	JWT session token; JWT token	JWT_Header + JWT_Payload + JWT_Signature	1.0 → Storage User	L0, L1
User_Record	user record	User_ID + Username + Password_Hash + Role	1.0 ↔ D1	L1
Folder_Command	Folder actions; JWT, folder cmds	JWT_Token + Folder_Action + (Folder_Name) + (Folder_ID) + (Parent_Folder_ID) * Folder_Action = [Create | Open | Back | Delete]	Storage User → 2.0	L0, L1
Folder_Record	folder records	{Folder_ID + Folder_Name + Parent_Folder_ID + User_ID}	2.0 ↔ D1	L1
Dashboard_View	Dashboard view	Current_Folder_ID + {Folder_Record} + {File_Summary} + (Empty_State_Flag)	2.0 → Storage User	L0, L1
Refresh	refresh	Current_Folder_ID	4.0 → 2.0	L1
File_Action	File actions, chunks	JWT_Token + [Binary_Chunk | Download_Request]	Storage User → System	L0
Binary_Chunk	binary chunks; Binary blocks; streamed blocks; multipart blocks	Upload_Session_ID + Chunk_Index + Chunk_Data + (Is_Final_Chunk)	Storage User → 3.0; 3.1 → 3.2 → 3.3; 3.0 → Telegram Bot API	L0, L1, L2
Folder_Check	folder check	Folder_ID + User_ID (returns Folder_Valid_Flag)	3.1 ↔ D1	L2
Total_File_Size	total file size	File_Size	3.1 → 3.4	L2
Transfer_Mode	≤ 50 MB: standard; > 50 MB: streaming	[Standard | Streaming] * Standard if File_Size ≤ 50 MB *	3.1 → 3.2	L2
Bytes_Pushed	bytes pushed	Bytes_Transferred	3.2 → 3.4	L2
Upload_Percent	progress %	Upload_Percent * 0 to 100 *	3.4 → Storage User	L2
Upload_Status	Upload % + errors; upload %, errors	Upload_Percent + (Error_Message)	3.0 → Storage User	L0, L1
Pause_Retry	pause + retry	Cool_Down_Seconds + Retry_Flag	3.3 → 3.2	L2
Connection_Error	connection error	Error_Message	3.3 → Storage User	L2
Telegram_Response	file_id; file_id, HTTP 429; 429, API status	[Telegram_File_ID | HTTP_429_Response] + API_Status	Telegram Bot API → 3.0 / 3.3	L0, L1, L2

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Data Flow	Alias	Composition	Source → Destination	Level
HTTP_429_Response	HTTP 429	Status_Code + Cool_Down_Seconds * Status_Code = 429 *	Telegram Bot API → 3.3	L0, L1, L2
File_Details	file_id, details; file_id + details; commit entry	Telegram_File_ID + File_Name + File_Size + Folder_ID + User_ID	3.0 / 3.3 → 4.0; 4.0 → D1	L1, L2
Search_Request	Search, UI prefs; search, prefs	JWT_Token + (Search_Text) + (Theme_Mode)	Storage User → 6.0	L0, L1
Name_Search	name search	User_ID + Search_Text	6.0 ↔ D1	L1
Matching_Records	matching records	{File_Record_ID + File_Name + File_Size + Folder_ID}	6.0 → Storage User	L1
Download_Request	download request; download request + JWT	JWT_Token + File_Record_ID	Storage User → 5.0 / 5.1	L1, L2
Ownership_Check	ownership, file_id; ownership check	User_ID + File_Record_ID + (Telegram_File_ID)	5.0 / 5.1 ↔ D1	L1, L2
Deny_Access_Error	deny access error	Error_Code + Error_Message	5.1 → Storage User	L2
Authorized_Request	authorized request	User_ID + File_Record_ID	5.1 → 5.2	L2
File_ID_Lookup	file_id lookup	File_Record_ID (returns Telegram_File_ID)	D1 → 5.2	L2
Internal_File_ID	internal file_id (never sent to browser)	Telegram_File_ID	5.2 → 5.3	L2
File_ID_Request	file_id request	Telegram_File_ID	5.0 / 5.3 → Telegram Bot API	L1, L2
Binary_Packet	binary packets	{Packet_Data}	Telegram Bot API → 5.0 / 5.3	L1, L2
Streamed_File	Streamed binary file; streamed file; streamed file response	{Binary_Packet} + (File_Name)	5.0 / 5.3 → Storage User	L0, L1, L2
Export_Request	Export request + admin credentials	Admin_Credentials + Export_Command	System Admin → 7.0	L0, L1
Schema_Dump	schema dump	{Table_Name + Table_Definition + {Row_Data}}	D1 → 7.0	L1
Backup_File	SQL backup file; backup file	Backup_File_Name + Schema_Dump	7.0 → System Admin	L0, L1

7. Data Structure Dictionary

Composite items that appear inside the data flows above.

Structure	Composition
Credentials	Username + Password
Admin_Credentials	Username + Password * must belong to a user with Role = "admin" *
Folder_Action	[Create | Open | Back | Delete]
Folder_Record	Folder_ID + Folder_Name + Parent_Folder_ID + User_ID
File_Summary	File_Record_ID + File_Name + File_Size + Upload_Date * Telegram_File_ID is excluded *

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Structure	Composition
Binary_Chunk	Upload_Session_ID + Chunk_Index + Chunk_Data + (Is_Final_Chunk)
Upload_Status	Upload_Percent + (Error_Message)
JWT_Token	JWT_Header + JWT_Payload + JWT_Signature
JWT_Payload	User_ID + Role + Expiry_Time * exact claims are TBD-007 *
HTTP_429_Response	Status_Code + Cool_Down_Seconds
Telegram_Response	[Telegram_File_ID | HTTP_429_Response] + API_Status
Export_Command	"export_metadata" * constant action name (proposed) *
Backup_File_Name	"metadata_backup_" + Export_Date + ".sql" * proposed *

8. Data Element Dictionary

Primitive data items (they cannot be broken down further). "Stored" shows whether the element is saved in D1.

Element	Type	Length / Domain	Stored	Meaning
User_ID	Integer	1 to 2^31-1	Yes	Unique identity of a user.
Username	String	3 to 50 chars	Yes	Unique login name.
Password	String	Policy TBD-006	No	Plain password received at registration or login; used only to create or compare a hash.
Password_Hash	String	≤ 255 chars	Yes	Hashed password.
Role	String	"user" | "admin"	Yes	Authorization level.
Created_At	DateTime	ISO 8601	Yes	Record creation time.
Folder_ID	Integer	≥ 1	Yes	Unique identity of a virtual folder.
Parent_Folder_ID	Integer	≥ 1 or NULL	Yes	Parent folder; NULL for root.
Current_Folder_ID	Integer	≥ 1 or NULL	No	Folder currently open in the UI; NULL means root.
Folder_Name	String	1 to 255 chars	Yes	Name of a virtual folder.
File_Record_ID	Integer	≥ 1	Yes	Internal identity of a file record.
Telegram_File_ID	String	≤ 255 chars	Yes	Persistent Telegram file identifier. Backend only.
File_Name	String	1 to 255 chars	Yes	Name of the uploaded file.
File_Size	Integer (64-bit)	0 to 2 GB	Yes	File size in bytes.
Upload_Date	DateTime	ISO 8601	Yes	Time the file was indexed.
Search_Text	String	1 to 255 chars	No	Characters typed in the search box.
Theme_Mode	String	"light" | "dark"	No	UI appearance choice. Where it is persisted is not defined in the SRS.
Upload_Session_ID	String	UUID	No	Identifies one upload in progress (proposed).
Chunk_Index	Integer	≥ 0	No	Position of a chunk in the upload.
Chunk_Data	Binary	Block size TBD	No	Raw bytes of one chunk. Streamed, never written to disk.
Is_Final_Chunk	Boolean	true | false	No	True on the last chunk; triggers file_id capture.
Bytes_Transferred	Integer	0 to File_Size	No	Bytes already forwarded to Telegram.
Upload_Percent	Integer	0 to 100	No	Bytes_Transferred ÷ File_Size × 100.
Transfer_Mode	String	"Standard" | "Streaming"	No	Standard if size ≤ 50 MB, otherwise streaming fallback.

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Element	Type	Length / Domain	Stored	Meaning
`Status_Code`	Integer	HTTP codes	No	HTTP status of a Telegram response (429 = throttled).
`Cool_Down_Seconds`	Integer	≥ 0	No	Wait time before retry after HTTP 429. Retry rules TBD-004.
`Retry_Flag`	Boolean	true | false	No	Whether the transfer will resume after cool-down.
`API_Status`	String	Free text	No	Status message returned by the Telegram API.
`Error_Code`	Integer	App-defined	No	Code for a controlled error (for example access denied).
`Error_Message`	String	≤ 255 chars	No	User-readable error text (network failure, size limit, access denied, folder not empty).
`Empty_State_Flag`	Boolean	true | false	No	True when the user has no files or folders (R5).
`Folder_Valid_Flag`	Boolean	true | false	No	True when the target folder exists and belongs to the user.
`JWT_Header`	String	Base64URL	No	Token type and signing algorithm (TBD-007).
`JWT_Signature`	String	Base64URL	No	Signature that proves the token is genuine.
`Expiry_Time`	DateTime	Epoch / ISO	No	Token expiry (TBD-007).
`Table_Name / Table_Definition / Row_Data`	String	SQL text	No	Parts of the exported metadata dump. Backup format is TBD.

9. Constants, Limits and Open Items

9.1 System Constants

Constant	Value	Source
Standard (non-chunked) upload ceiling	50 MB	SRS 2.5, FR-016
Standard download limit	20 MB	SRS 2.5
Absolute maximum file size	2 GB (via streaming/chunking)	SRS 2.5, FR-017; Decision Table R13
Throttling status code	HTTP 429	FR-022, R15

9.2 Items Marked TBD or Proposed

- Password policy and hashing algorithm (TBD-006); JWT algorithm, expiry and claims (TBD-007).
- Administrator role model (TBD-009); the Role column in Users is a proposed way to support it.
- SQL vendor, final schema names and indexes (TBD-003); data types above are logical and may be adjusted.
- Retry counts, intervals and chunk size (TBD-004).
- Columns Created_At, Upload_Date and Upload_Session_ID are proposed additions that are not named in the SRS.

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